Appendix C

Glossary

the words, in one line each, and where to read more

114 terms, each with the chapters that teach it (grey numbers on the right). One line is a reminder, not an explanation: if a definition doesn't click, read the chapter, then the vault notes listed in its deep-dive box.

A

abstract base class	a class that declares methods its subclasses must provide; it can't be instantiated itself	5
adjacency list	a graph stored as a dict from each node to the list of its neighbours	21
aliasing	two names referring to one object, so a change made through one is seen through the other	3
amortised cost	the average cost per operation over a long run: `list.append` is O(1) amortised despite occasional resizes	20
argv	the list of command-line arguments; `sys.argv[0]` is the program itself	14
atomic write	write a temporary file, then `os.replace` it, so readers see the old file or the new one, never half	8
aware datetime	a datetime that carries a time zone; a naive one doesn't, and the two can't be compared	11

B

base case	the input a recursive function answers directly, without recursing	20
base64	an encoding of bytes as ASCII text; anyone can decode it, so it is not encryption	13
big-O	how cost grows with input size, ignoring constant factors: O(1), O(log n), O(n), O(n log n), O(n²)	20
binary search	find a position in sorted data by halving the range each step: O(log n)	10 · 20
breadth-first search	explore a graph in rings of distance using a queue; finds the fewest edges	21
broadcasting	NumPy's rule for combining arrays of different shapes by stretching size-1 axes	18
bytecode	the instructions CPython compiles source to and its virtual machine runs (cached in .pyc files)	2
bytes	an immutable sequence of integers 0-255: what files, sockets and hashes really hold	8 · 13

C

call stack	the stack of frames for the calls in progress; each call pushes a frame, each return pops it	6 · 20
closure	a function that remembers variables from the scope where it was defined	4

code point	the number Unicode assigns to a character; a `str` is a sequence of code points	9
comprehension	an expression building a list, dict or set from a loop: `[x * x for x in xs if x]`	1
context manager	an object used with `with` whose `__exit__` runs however the block ends	6
coroutine	an `async def` function's object; it runs until an `await`, letting the event loop run others	16
cursor	the object that executes SQL and walks through the result rows	12
cycle collector	the part of the garbage collector that frees groups of objects referring only to each other	3

D

dataclass	a class whose `__init__`, `__repr__` and `__eq__` are generated from its field declarations	5
decorator	a function that takes a function and returns a replacement, applied with `@name`	4
deep copy	a copy of an object and, recursively, everything it refers to (`copy.deepcopy`)	3
depth-first search	explore a graph as far as possible along each branch before backing up, using a stack	21
dict	a hash table from keys to values; O(1) lookup on average, remembers insertion order	1 · 10
Dijkstra's algorithm	cheapest paths from one node when weights are non-negative, using a heap	21
duck typing	using an object for what it can do (its methods), not for what class it is	5
dunder method	a `__double_underscore__` method Python calls for syntax: `__len__`, `__eq__`, `__iter__`	5
dynamic programming	solve each overlapping subproblem once and reuse the answer: memoised recursion or a table	20 · 21

E

EAFP	'easier to ask forgiveness than permission': try the operation and handle the exception	6
edit distance	the fewest single-character inserts, deletes and replacements turning one string into another	21
encoding	the rule turning text into bytes and back; UTF-8 unless you have a reason	8 · 13
environment variable	a name-value string a process inherits from its parent (`os.environ`)	14
event loop	asyncio's scheduler: runs ready tasks, and switches at every `await` that has to wait	16
exception	an object raised to signal an error; it unwinds the stack until an `except` matches	6
exit code	the number a process returns to the OS: 0 for success, anything else for failure	14

F

f-string	a string literal with `{expressions}` and format specs: `f"{total:,.2f}"`	9
foreign key	a column referring to another table's primary key, so rows link by id	12

framing	marking where each message ends in a byte stream such as TCP (lines, length prefixes)	15
free-threaded build	a CPython build without the GIL (3.13 onwards), so threads can run Python in parallel	16

G

generator	a function with `yield`; calling it returns an iterator that runs the body lazily	4
GIL	the global interpreter lock: on the default build, one thread runs Python bytecode at a time	16
graph	nodes joined by edges, directed or not, possibly weighted	21
greedy algorithm	always take the best-looking choice now; correct only when that choice can be proved safe	21

H

hash function	a function from data to a fixed-size digest; cryptographic ones make collisions infeasible	13
hashable	has a hash that never changes, so it can be a dict key or set member	3 · 5
heap	a tree stored in a list where every parent is <= its children; `heapq` pops the smallest in O(log n)	10 · 21
HMAC	a tag made from a message and a secret key; proves the message is intact and from a key holder	13
HTTP	the text request-response protocol of the web: method, path, headers, body, status code	15

I

immutable	can't be changed after creation (int, str, tuple, frozenset); 'changes' make new objects	3
index (database)	a sorted structure (a B-tree) that lets a query find rows without scanning the table	12
interpreter	the program that runs Python code: `python3` is CPython's interpreter	14 · 16
invariant	a statement that is true before and after every step of a loop or method	20
iterator	an object with `__next__` that hands out values one at a time; iterables make iterators	4

J

join	an SQL query combining rows of two tables where a condition (usually key = foreign key) holds	12
JSON	a text format for nested lists, dicts, strings, numbers, booleans and null	8

L

LEGB	the order Python looks names up: Local, Enclosing, Global, Built-in	4
library card	this book's six questions for learning any library	7
linked list	nodes each holding a value and a reference to the next; O(1) insert at a known node, O(n) indexing	20
lock	lets one thread at a time into a critical section (`threading.Lock`)	16
lock file	exact versions and hashes of every dependency, so installs are reproducible	19

M

memoisation	caching a function's results by argument (`functools.cache`)	20 · 21
merge sort	split in half, sort each half, merge: O(n log n), stable	20
method resolution order	the order classes are searched for an attribute (C3 linearisation; `Class.__mro__`)	5
mock	a stand-in object that records calls and returns canned answers in a test	17
module	a `.py` file (or compiled extension) loaded by `import`; a package is a folder of modules	2 · 7
mutable	can be changed in place (list, dict, set, most instances)	3

N

ndarray	NumPy's n-dimensional array: one typed buffer plus a header of dtype, shape and strides	18
normalisation	Unicode: one canonical form for equal-looking text (NFC). Databases: each fact stored once	9 · 12

P

parameter binding	passing values to SQL through `?` placeholders, never by building strings: stops injection	12
passport	this book's twelve questions for understanding any programming language	0
pickle	Python's object serialiser; loading a pickle can run arbitrary code, so never load untrusted ones	8
pip	the installer that resolves, downloads, verifies and unpacks packages into an environment	19
process	a running program with its own memory; processes don't share objects	14 · 16
protocol	a set of methods a type must have to be used somewhere (iteration, `typing.Protocol`)	5 · 17
pseudo-random	numbers from a deterministic generator: reproducible with a seed, predictable, so not for secrets	11

R

race condition	a bug where the result depends on how threads' steps interleave	16
recursion	a function solving a problem by calling itself on smaller inputs	20
reference count	how many references point at an object; at zero CPython frees it at once	3
regular expression	a pattern language for matching text (`re`)	9

S

salt	random bytes stored with a password hash so equal passwords hash differently	13
scope	the region of code where a name is visible	4
secrets	the module for unguessable random values (tokens, passwords); `random` is not safe for these	11 · 13
shallow copy	a new container holding the same element objects (`list(xs)`, `xs.copy()`, `copy.copy`)	3
signal	an asynchronous notification from the OS to a process (SIGINT on Ctrl-C, SIGTERM)	14
socket	an endpoint for network communication; TCP sockets carry a byte stream	15

spike	a throwaway 20-line program written to answer one question about a library	7
SQL injection	attacker text ending a string inside an SQL statement and adding its own commands	12
stable sort	a sort that keeps equal items in their original order (Python's always is)	20
stack	last in, first out; a Python list with `append` and `pop`	21
stderr	the stream for messages and errors; stdout is for results, so pipes stay clean	14
stride	how many bytes to step in memory to move one place along an array axis	18
subprocess	running another program from Python and collecting its output and exit code	14
supply-chain attack	attacking users through a dependency: typosquatted names, hijacked or malicious packages	19

T

TCP	the reliable, ordered byte-stream protocol under HTTP, SMTP and most of the internet	15
thread	a line of execution inside a process; threads share memory	16
timing attack	learning a secret from how long a comparison takes; prevented by `hmac.compare_digest`	13
Timsort	Python's sort: a merge sort that exploits runs already in order	20
TLS	encryption and server authentication for a TCP connection (the S in HTTPS)	15
topological order	an order of a directed graph's nodes where every edge points forward; exists only without cycles	21
traceback	the printed stack of calls that led to an exception, oldest first	6
transaction	a group of changes that commit together or not at all	12
tree	nodes with children and no cycles; in a binary search tree, left < node < right	20
type hint	an annotation like `def f(x: int) -> str`; checked by tools, ignored when running	17

U

Unicode	the standard numbering every character in every script	9
unit test	a test of one small piece of code in isolation	17
UTC	Coordinated Universal Time; store and compute in UTC, convert to local time for display	11

V

vectorisation	replacing Python loops with whole-array operations that run in compiled code	18
view	an array sharing another array's buffer: changing one changes the other	18
virtual environment	a folder with its own site-packages so each project has its own dependencies	2 · 19

W

wheel	a built package file (.whl) that pip unpacks without running any setup code	17 · 19